

# Mohith Munagala

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## PROFESSIONAL SUMMARY

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Cloud & DevOps Engineer with 5+ years of experience designing and managing **scalable cloud infrastructure across AWS and Azure**, with strong expertise in **Kubernetes (EKS, AKS), Docker, and microservices architecture**. Proven experience building and automating **CI/CD pipelines using GitHub Actions, Jenkins, GitLab CI/CD, and Azure DevOps**, enabling reliable and efficient deployments. Skilled in **Infrastructure as Code (Terraform, CloudFormation, Helm)** and implementing secure **cloud networking, IAM, RBAC, and Key Vault/KMS solutions**. Hands-on experience in developing **high-performance platforms supporting AI/ML workloads (LLMs, RAG) on GPU-enabled Kubernetes environments**. Strong background in **observability (Prometheus, Grafana, CloudWatch, Splunk)**, **automation (Python, Bash)**, and **DevSecOps practices**, delivering highly available, secure, and production-ready systems.

## EDUCATION

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**Eastern Illinois University – Charleston, Illinois**

Aug 2023 – May 2025

MS in Computer Science

**Relevant Coursework:** Cloud Computing, DevOps, Infrastructure as Code (IaC), CI/CD, Kubernetes, Scripting, Cloud Architecture, Security, AI Models and Automation Tools.

## TECHNICAL SKILLS

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**Cloud Platforms:** AWS (EC2, S3, RDS, Lambda, VPC, IAM, KMS, CloudWatch, Route 53, API Gateway, SQS, EKS, DynamoDB), Microsoft Azure (AKS, App Services, Key Vault, Azure AD, ARM Templates)

**DevOps & CI/CD:** GitHub Actions, Jenkins, Azure DevOps, GitLab CI/CD, Argo CD (GitOps), AWS CodeDeploy

**Infrastructure as Code (IaC):** Terraform, AWS CloudFormation, Azure Bicep, AWS CDK, Helm

**Containers & Orchestration:** Docker, Kubernetes (EKS, AKS, On-Prem), OpenShift, Traefik Ingress Controller

**Monitoring & Observability:** Prometheus, Grafana, Amazon CloudWatch, SonarQube, Azure Monitor, Splunk, Datadog

**Scripting & Automation:** Python, Bash, Ansible, PowerShell, YAML, JSON

**Security & Compliance:** AWS IAM, Azure RBAC, KMS, Azure Key Vault, OAuth 2.0, S3 Bucket Policies, VPC Security Groups, HIPAA, GDPR

**Networking:** VPC Architecture, Subnets, Route Tables, NAT Gateway, Load Balancers (ALB/NLB), DNS (Route 53), SSL/TLS

**Databases & Storage:** PostgreSQL, MySQL, DynamoDB, Azure SQL, Elasticsearch, Neo4j

**AI / ML & GPU Platforms:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Knowledge Graphs, Ollama, GPU Infrastructure, NVIDIA H200

## EXPERIENCE

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**Caterpillar**  
Cloud Engineer

**Peoria, IL**  
Sep 2025 – Present

- Led design and deployment of **GPU-enabled Kubernetes (On-Prem) infrastructure** for enterprise AI platform, supporting **LLM inference workloads** with high availability and scalability.
- Architected and implemented **RAG (Retrieval-Augmented Generation) pipeline** integrating **LLMs, PostgreSQL, and Neo4j**, enabling context-aware AI applications.
- Deployed and managed **containerized applications using Docker and Kubernetes**, ensuring scalable **microservices architecture** with autoscaling and self-healing.
- Built and managed **Docker container images** for microservices and AI workloads, enabling consistent and portable deployments across **Kubernetes (on-prem clusters)**.
- Integrated **Docker image build and deployment workflows into CI/CD pipelines using GitHub Actions and Argo CD**, including image versioning and secure container registry management.
- Implemented **CI/CD pipelines using GitHub Actions and Argo CD (GitOps)** for automated build, test, and deployment across multiple environments.

- Designed and optimized **Kubernetes resource scheduling and GPU utilization**, improving application performance and infrastructure efficiency.
- Configured **secure cloud and on-prem authentication using Azure AD, RBAC, and Azure Key Vault**, ensuring compliance with enterprise security standards.
- Built and maintained **infrastructure automation and deployment workflows**, improving release reliability and reducing manual effort.
- Developed and maintained **Ansible playbooks** and roles for node provisioning, configuration management, and patch automation.
- Designed **ingress and networking architecture using Traefik, SSL/TLS, and load balancing**, ensuring secure service exposure.
- Implemented **distributed storage solutions using Longhorn** for persistent and highly available data storage across Kubernetes clusters.
- Established **monitoring and observability using Prometheus, Grafana, and Azure Monitor**, enabling proactive alerting and performance tracking.
- Built and managed **GitHub self-hosted runners** to support secure and scalable CI/CD pipelines for Kubernetes workloads.
- Conducted **performance tuning, capacity planning, and system optimization** for high-throughput AI applications.
- Performed **root cause analysis (RCA)** across **Kubernetes, networking, container runtime, and database layers** to ensure production stability.
- Improved **system reliability, scalability, and production performance** using DevOps and cloud automation best practices.
- Collaborated with cross-functional teams to implement **DevOps best practices, automation, and cloud-native solutions**.



**British Airways**  
**Cloud Engineer**

**New York, USA**  
**June 2024 – Sep 2025**

- Designed and deployed **AWS EKS-based Kubernetes infrastructure** using **Auto Scaling and ALB/NLB**, supporting high-traffic, production-grade flight booking applications.
- Provisioned and managed **cloud infrastructure using Terraform, AWS CloudFormation, and Helm**, enabling automated and scalable **Infrastructure as Code (IaC)** deployments.
- Built and secured **AWS VPC architecture (subnets, route tables, NAT Gateway, IAM, KMS)** ensuring compliance with enterprise security and governance standards.
- Built and optimized **CI/CD pipelines using GitHub Actions and Jenkins** for automated build, test, and deployment of **40+ microservices**.
- Implemented **Docker containerization and Kubernetes orchestration (EKS)** to deploy scalable, highly available **microservices architecture**.
- Designed and deployed **REST APIs using AWS API Gateway and AWS Lambda**, enabling serverless and event-driven application workflows.
- Developed **Python (Boto3) automation scripts** for **EKS scaling, S3 audits, and CloudWatch log management**, improving operational efficiency and reducing manual tasks.
- Implemented **blue/green and canary deployment strategies using Kubernetes and Helm**, enabling zero-downtime releases and improved deployment reliability.
- Configured **monitoring and observability using Prometheus, Grafana, AWS CloudWatch, and Splunk**, ensuring proactive alerting and system performance monitoring.
- Performed **performance testing and load testing using JMeter and Locust**, optimizing autoscaling and improving application performance under dynamic workloads.
- Integrated **SonarQube** into **CI/CD pipelines (Jenkins, GitHub Actions)** for **static code analysis**, enforcing **quality gates**, detecting **code smells, bugs, and security vulnerabilities**, and improving **code quality and secure deployments**.
- Designed and implemented **high availability and disaster recovery (DR) solutions across multi-AZ using Route 53 and CloudFormation failover mechanisms**.
- Built and maintained **secure SOAP APIs for IATA integrations**, ensuring compliance with airline data standards and external system interoperability.

- Collaborated with QA and Dev teams to implement **API testing using Postman, SoapUI, and automation scripts**, improving release quality and reducing defects.
  - Participated in **Agile/Scrum processes**, including sprint planning, release coordination, and cross-team collaboration for continuous delivery.
  - Applied **DevOps best practices, automation, and cloud-native architecture patterns** to improve system reliability, scalability, and deployment efficiency.
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**TCS**  
**Cloud Engineer**

**Hyderabad, India**  
**May 2021 – July 2023**

- Provisioned and managed **AWS cloud infrastructure (EC2, VPC, S3, RDS, Lambda, IAM)** supporting scalable and secure financial applications.
- Automated **Infrastructure as Code (IaC)** using **Terraform and AWS CloudFormation**, enabling consistent deployments across **dev, UAT, and production environments**.
- Built and optimized **CI/CD pipelines using GitLab CI and AWS CodeDeploy** for automated build, test, and deployment of **containerized microservices**.
- Implemented **Docker containerization and Kubernetes orchestration (Amazon EKS)** to deploy scalable, highly available **microservices architecture**.
- Deployed and managed **microservices on Kubernetes (EKS)** using **Helm charts**, improving deployment automation and release consistency.
- Developed **serverless applications using AWS Lambda, API Gateway, and SQS** to support real-time payment processing systems.
- Created **Python and Bash automation scripts** for EC2 lifecycle management, AMI backups, log cleanup, and CloudWatch monitoring integration.
- Designed and enforced **AWS security best practices using IAM policies, KMS encryption, and S3 bucket policies**, ensuring compliance with enterprise standards.
- Configured **monitoring and observability using Prometheus, Grafana, and AWS CloudWatch**, enabling proactive alerting and performance monitoring.
- Performed **root cause analysis (RCA) and incident management using CloudTrail, VPC Flow Logs, and Splunk**, improving system reliability and uptime.
- Managed **API lifecycle (REST and SOAP APIs)** including versioning, authentication, and schema validation using cloud-native services.
- Maintained **runbooks, documentation, and automation repositories in Git**, supporting operational efficiency and team onboarding.
- Collaborated with cross-functional teams to implement **DevOps and DevSecOps practices**, improving deployment automation and system security.
- Participated in **Agile/Scrum processes**, delivering sprint-based cloud and DevOps solutions.
- Implemented and enforced **compliance and governance policies (HIPAA, GDPR)** in collaboration with InfoSec teams.

## **CERTIFICATIONS**

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- AWS Certified Solution Architect
- Microsoft Azure Fundamentals
- AWS Certified Cloud Practitioner
- Certified Kubernetes
- Certified Terraform